

Software Development Internship at Flipkart

BACHELOR OF TECHNOLOGY

SOFTWARE DEVELOPMENT INTERN, FLIPKART



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1. Introduction

This report documents the work carried out during my Software Development Internship at Cogniease Techno Labs Pvt. Ltd. from August 2026 to September 2026. The internship was undertaken as part of the Bachelor of Technology curriculum at the Indian Institute of Information Technology, Allahabad.

During this internship, I worked as a Full-Stack Developer on "Taru" — an Emotional Wellbeing Platform designed for college students. Taru aims to address the growing mental health crisis among students by providing a single, confidential digital space that combines self-assessment tools, peer support, professional counseling access, and stress-relief activities.

The internship provided hands-on experience in modern web development technologies including React.js, Node.js, Express.js, MongoDB, Socket.IO, and WebRTC. Over the course of the internship, I contributed 33 commits to the project repository, building features that span the entire stack of the application — from database schema design and REST API development to real-time game engines and responsive frontend interfaces.

1.1 Objectives

- Gain practical experience in full-stack web application development in a professional environment.
- Understand the software development lifecycle (SDLC) in a startup setting.
- Develop proficiency in real-time communication systems such as WebSockets and WebRTC.
- Learn database design, RESTful API architecture, and multi-role authentication systems.
- Contribute meaningfully to a production-grade application that serves real users.

2. About the Organization

Cogniease Techno Labs Pvt. Ltd. is a technology company focused on building digital solutions for mental health and emotional wellbeing. The company is driven by the mission of making mental health support more accessible, affordable, and stigma-free — particularly for young adults and college students who are among the most affected demographics.

The company's flagship product is Taru, a platform that brings together evidence-based mental health assessments, anonymous peer support, professional counseling, mood tracking, and interactive stress-relief tools under a single roof. The development team follows Agile practices with GitHub-based code reviews, feature branching, and pull request workflows.

3. Problem Statement

Mental health issues among college students have reached alarming levels.

Research indicates that approximately one in three college students experiences significant anxiety or depression during their academic career. Despite this prevalence, the majority of affected students never seek professional help. The key barriers include:

- **Stigma:** Fear of being judged by peers, faculty, or family members.
- **Accessibility:** Long waiting lists at campus counseling centers and limited availability of professionals.
- **Cost:** Professional mental health services are often prohibitively expensive for students.
- **Fragmentation:** Existing solutions address only one aspect (e.g., only meditation or only therapy booking) without providing a holistic platform.

Taru was conceived as a unified platform that addresses all these barriers simultaneously. It offers daily check-ins, peer chat, psychiatrist video consultations, AI chatbot support, mood tracking, and stress-relief games — all free for students and within a single confidential application.

4. Project Overview — Taru Platform

Taru is a full-stack web application that serves three distinct user roles, each with a dedicated portal, authentication flow, and dashboard. The platform is designed to be student-first while also empowering mental health professionals and educational institutions with the tools they need.

4.1 Student Portal

The primary interface for students provides a comprehensive suite of wellbeing tools. Students can perform daily check-ins that assess seven dimensions of emotional health (mood, energy, stress, sleep, concentration, social support, and motivation). The platform tracks check-in streaks to encourage consistent self-monitoring. Students can visualize mood trends through weekly charts and a calendar-based history view.

For support, students can connect through anonymous peer chat (random matchmaking with auto-generated usernames), an AI chatbot for 24/7 guidance, or by booking appointments with certified psychiatrists for WebRTC-based video consultations. Stress-relief features include a Breathing Bubble component and five multiplayer games designed to ground and calm the mind.

4.2 Psychiatrist Portal

A dedicated portal for mental health professionals includes its own registration and login system, a dashboard to manage upcoming appointments, tools to configure weekly availability slots, and the ability to conduct video consultations with students. During sessions, psychiatrists can view the student's mood history and use an in-meeting text chat panel for sharing resources.

4.3 Institution Portal

Educational institutions get access to an admin dashboard with aggregate student analytics, visual mood trend charts across the student body, an enrolled student list with search and filtering capabilities, and check-in participation rate monitoring. This portal enables data-driven decisions for campus-wide wellbeing programs.

5. Technology Stack

The platform uses a modern JavaScript/TypeScript stack chosen for its suitability for real-time applications, rapid UI development, and strong ecosystem support.

5.1 Frontend

Technology	Version	Purpose
React	19.0	UI library with hooks and functional components
TypeScript	5.7+	Type-safe development for reliability
Vite	6.0	Fast build tool and development server
Tailwind CSS	4.0	Utility-first CSS framework for rapid styling
React Router	v7.18	Client-side routing and navigation
TanStack React Query	5.x	Server state management and API data caching
Lucide React	1.x	Consistent, customizable icon library
Socket.IO Client	4.8	Real-time WebSocket communication

5.2 Backend

Technology	Version	Purpose
Node.js	≥18.0	JavaScript runtime for server-side logic
Express.js	4.21	Minimal web framework for REST API endpoints
MongoDB + Mongoose	8.14	NoSQL database with Object Data Modeling

Socket.IO	4.8	Real-time bidirectional event-based communication
JSON Web Tokens	9.0	Stateless authentication tokens
bcryptjs	2.4	Secure password hashing (salt factor 12)
WebRTC	—	Peer-to-peer video/audio communication
Express Validator	7.2	Server-side input validation middleware

6. System Architecture

Taru follows a three-tier client-server architecture that supports both RESTful HTTP communication and real-time bidirectional communication via WebSockets. The codebase is organized into two top-level directories — `Taru_frontend` and `Taru_backend` — each with a clear internal structure.

6.1 Client Layer

The frontend is built with React 19 and TypeScript, bundled by Vite, and styled with Tailwind CSS 4. It contains three separate portal interfaces, each guarded by its own authentication context (`AuthContext`, `PsychiatristAuthContext`, `InstitutionAuthContext`) and protected route components. React Router v7 handles all client-side navigation, while TanStack React Query manages server state, caching, and background data refetching.

6.2 Application Layer

The backend is a Node.js server built with Express.js, organized into controllers (9 modules covering auth, users, appointments, availability, check-ins, chatbot, meetings, psychiatrists, and institutions), models (7 Mongoose schemas), routes, and authentication middleware. A dedicated `realtime/` directory houses the Socket.IO server, matchmaking engine, room management, and game logic modules.

6.3 Data Layer

MongoDB serves as the primary database, accessed through Mongoose ODM. The seven collections are `User`, `DailyCheckIn`, `Appointment`, `Availability`, `Psychiatrist`, `Institution`, and `Meeting`. Schemas include pre-validation hooks (e.g., auto-computing `totalScore` in `DailyCheckIn` from seven metric scores), pre-save middleware (password hashing with `bcrypt`), and compound indexes for query optimization.

6.4 Real-Time Module

The real-time layer uses Socket.IO for WebSocket communication and WebRTC for peer-to-peer video. It features an event-driven architecture with in-memory room management, FIFO queue-based matchmaking per game type, a generalized game

module lookup pattern (allowing new games to be plugged in), and WebRTC signaling relay for ICE/SDP exchange. Five multiplayer games are implemented: Memory, Connect Four, Beach Balls (with a custom physics engine), Color Wars (with powerups), and Sea Battle.

7. Features Implemented

The following major features were designed, developed, and deployed during the internship:

- **Multi-Role Authentication:** Three separate JWT-based auth flows for Students, Psychiatrists, and Institutions, each with bcrypt password hashing, HTTP-only cookie sessions, and dedicated React context providers.
- **Daily Check-in System:** Science-backed assessments tracking seven wellbeing dimensions with auto-computed total score, streak tracking, weekly mood charts, and calendar-based history.
- **WebRTC Video Conferencing:** Custom useWebRTC React hook managing RTCPeerConnection lifecycle, ICE candidate exchange, SDP offer/answer via Socket.IO signaling, and an in-meeting text chat panel.
- **Real-Time Multiplayer Games:** Five games with server-authoritative logic — Memory (card matching), Connect Four (animated drops), Beach Balls (physics engine with collisions), Color Wars (powerups, virtual joystick), and Sea Battle (fleet placement, turn-based combat).
- **Anonymous Peer Chat:** Random matchmaking with auto-generated usernames (taru_XXXXXXX) enabling anonymous student-to-student support conversations.
- **AI Chatbot Integration:** Taru AI chatbot providing 24/7 mental health guidance, coping strategies, and conversational support for students.
- **Stress Relief Activities:** Breathing Bubble component with animated visual guidance and a curated set of interactive mindfulness exercises.

8. My Contributions

Over 33 commits (August 9 – September 19, 2026), the following work was delivered:

Area	Work Delivered
Backend Setup	Express server, MongoDB connection, 7

	Mongoose schemas, API client, environment config
Psychiatrist Portal	Full-stack: registration, authentication, dashboard, availability, appointments, profile
Institution Portal	Full-stack: registration, auth, student analytics dashboard, mood trends, student list
Video Conferencing	WebRTC peer-to-peer calls with Socket.IO signaling, appointment integration, meeting chat
Multiplayer Games	5 complete games with backend logic engines and animated React frontend components
Peer Chat & AI	Anonymous random chat with matchmaking + Taru AI chatbot integration
Check-in & Mood	7-dimension daily assessments, total score computation, streak tracking, mood history
UI/UX & Landing	Responsive landing page, breathing bubble, stress-relief view, profile management

9. Challenges Faced & Solutions

Building a production-grade platform with real-time features presented several technical challenges. The key ones and their solutions are described below.

9.1 WebRTC Connectivity

Establishing reliable peer-to-peer video connections was challenging because ICE candidate exchange and SDP offer/answer must be timed precisely. The solution involved implementing a proper signaling flow via Socket.IO with STUN server configuration for NAT traversal, ensuring connections work across different networks.

9.2 Real-Time Game Synchronization

Keeping game state consistent across two players with variable network latency could lead to desynchronization. This was solved by adopting a server-authoritative model where all game logic and move validation runs on the server; clients receive only authoritative state updates and render accordingly.

9.3 Multi-Role Authentication

Supporting three user types with different schemas and auth requirements needed careful architectural planning. The solution was to design three independent authentication contexts on the frontend and separate middleware chains on the backend, each with its own JWT secret and cookie name.

9.4 CORS & Production Deployment

Cross-origin issues arose when deploying the frontend and backend to different domains. This was resolved by simplifying the CORS configuration to accept an array of allowed origins from environment variables, enabling seamless deployment on Vercel.

9.5 Physics Engine for Beach Balls

Implementing realistic ball physics with collision detection in a real-time multiplayer setting was complex. A custom physics module was built with velocity vectors, boundary

bouncing, and inter-ball collision calculations, all running at a fixed server tick rate and synchronized to both clients.

10. Key Learnings & Conclusion

10.1 Key Learnings

- **Full-Stack Development:** End-to-end proficiency with React, Node.js, Express, and MongoDB in a production environment.
- **Real-Time Systems:** Socket.IO architecture, event-driven design, room management, and matchmaking queues.
- **WebRTC:** Peer-to-peer video communication with ICE framework, SDP negotiation, and STUN/TURN servers.
- **Authentication & Security:** Multi-role JWT auth, bcrypt hashing, cookie sessions, and protected route patterns.
- **Game Development:** Physics simulation, turn-based logic, server-authoritative state, and frontend animations.
- **Database Design:** Mongoose schemas with pre-hooks, compound indexes, and relationship modeling.
- **Deployment:** Vercel deployment, environment management, CORS handling, and production debugging.
- **Professional Practices:** Git workflows (branching, PRs, code reviews), Agile methodology, and documentation.

10.2 Conclusion

The internship at Cogniease Techno Labs was a highly valuable experience that bridged the gap between academic knowledge and real-world software engineering. Working on Taru gave me the opportunity to build meaningful technology that directly addresses the mental health challenges faced by college students.

Over the internship, I successfully delivered the Psychiatrist and Institution portals, a WebRTC video conferencing system, five real-time multiplayer games, anonymous peer chat, and AI chatbot integration. These contributions strengthened my technical skills across the full stack and prepared me well for a career in software engineering.

10.3 Future Scope

- AI-powered mood prediction and early intervention alerts for at-risk students.
- Redis-backed horizontal scaling to replace in-memory room storage for production loads.
- React Native mobile application for on-the-go check-ins and mood tracking.
- Multi-party WebRTC for group therapy sessions.
- Push notifications for appointment reminders, streak alerts, and check-in nudges.